Biomedical Signal Processing and Control 69 (2021) 102923

Contents lists available at ScienceDirect

Biomedical Signal Processing and Control

journal homepage: www.elsevier.com/locate/bspc

Deep learning and regression approaches to forecasting blood glucose levels for type 1 diabetes

Meng Zhang, Kevin B. Flores, Hien T. Tran *

Center for Research in Scientific Computation, Department of Mathematics, North Carolina State University, Raleigh, NC 27695, USA

ARTICLE INFO

Keywords:
Neural networks
Regression
Encoder decoder
Time series forecasting
Diabetes

ABSTRACT

Objective: Controlling blood glucose in the euglycemic range is the main goal of developing the closed-loop insulin delivery system for type 1 diabetes patients. The closed-loop system delivers the amount of insulin dose determined by glucose predictions through the use of computational algorithms. A computationally efficient and accurate model that can capture the physiological nonlinear dynamics is critical for developing an efficient closed-loop system. Methods: Four data-driven models are compared, including different neural network architectures, a reservoir computing model, and a novel linear regression approach. Model predictions are evaluated over continuous 30 and 60 min time horizons using real-world data from wearable sensor measurements, a continuous glucose monitor, and self-reported events through mobile applications. The four data-driven models are trained on 12 data contributors for around 32 days, 8 days of data are used for validation, with an additional 10 days of data for out-of-sample testing. Model performance was evaluated by the root mean squared error and the mean absolute error. Results: A neural network model using an encoder-decoder architecture has the most stable performance and is able to recover missing dynamics in short time intervals. Regression models performed better at long-time prediction horizons (i.e., 60 min) and with lower computational costs. Significance: The performance of several distinct models was tested for individual-level data from a type 1 diabetes data set. These results may enable a feasible solution with low computational cost for the time-dependent adjustment of artificial pancreas for diabetes patients.

1. Introduction

Type 1 Diabetes Mellitus (T1D) is a chronic disease with an incidence of 15 per 100,000 people worldwide between 1980 and 2019 [1]. Due to the destruction of insulin-producing pancreatic β cells in T1D patients, the pancreas can produce little or no insulin. Disruption of insulin regulation can lead to a diseased state since this hormone helps regulate the storage and utilization of glucose obtained from carbohydrate intake. Lack of insulin will lead to a failure to control regular glucose-insulin transformation and can cause long-term complications including permanent damage to organs and nerves. There is no cure for T1D, patients with this disease need lifetime exogenous insulin injections to manage Blood Glucose (BG) levels in the euglycemic range, i.e., 70–140 mg/dl with an empty stomach and below 200 mg/dl after 2 h of digestion [2].

In recent years, insulin pumps and Continuous Glucose Monitoring (CGM) are two technologies that have been used to develop sensor-augmented pump therapy for T1D treatment [3]. Researchers try to combine CGM and insulin pumps with a computer algorithm to compose a closed-loop system, i.e., artificial pancreas. If the CGM goes above a certain threshold then insulin starts being delivered through the pump and stops if BG levels are sufficiently low. A key challenge to developing this system is designing an accurate control algorithm that predicts the trend of glucose levels and manages the insulin amount to infuse. For this reason, the development of models that can accurately predict BG levels is significant to the effectiveness of the artificial pancreas.

Previously, three types of models have been developed for blood glucose prediction: physiological models, data-driven models, and hybrid models. Physiological models are mathematical models that apply the biological understanding of glucose transformation to simulate the dynamics of the insulin-glucose regulation system. Data-driven models include statistical models (e.g., linear regression) and artificial neural networks (e.g., deep learning models). Finally, hybrid models combine physiological models with data-driven models.

In the framework of physiological models, the minimal model by Bergman et al. [4] describes the metabolism of glucose and insulin

* Corresponding author.
E-mail addresses: mzhang22@ncsu.edu (M. Zhang), kbflores@ncsu.edu (K.B. Flores), tran@ncsu.edu (H.T. Tran).

https://doi.org/10.1016/j.bspc.2021.102923
Received 12 February 2021; Received in revised form 21 June 2021; Accepted 24 June 2021
Available online 8 July 2021
1746-8094/© 2021 Elsevier Ltd. All rights reserved.

regulation. Man et al. [5] proposed a model that consists of glucose-insulin systems with the effect of glucose absorption in gastro-intestinal, endogenous glucose production, glucose utilization, and renal excretion. Later, Man et al. extended this physiological model to include effects from physical activity [6].

With regards to the data-driven models, a complete review can be found in [7]. In recent years, models attempting to predict BG levels have included Auto-Regression (AR) models [8–13], XG-Boost [14], Grammatical Evolution (GE) [15], Genetic Programming (GP) [16], Neural Basis Expansion for Interpretable Time-Series Forecasting (N-BEATS) [17], Latent-Variable (LV) based models [18], Generative Adversarial Networks (GANs) [19], shadow network and error imputation [20], Convolutional Neural Networks (CNNs) [21], and Recurrent Neural Networks (RNNs) [22–34].

Hybrid models are usually designed to replace or add features simulated by the physiological model in data-driven algorithms. In [35], Bertachi et al. used a mathematical model to construct additional features (insulin on board, carbohydrate on board and activity on board) which are then used as inputs to artificial neural networks. Zecchin et al. [36] considered the carbohydrates absorption model as an input feature in the jump neural network to improve predictive accuracy.

The main goal of this study is to design data-driven personalized BG predictive models for T1D that are computationally efficient, suitable for wearable devices, and perform well on common problems that arise in the analysis of real-world data (missing data, uncalibrated data) for insulin therapy. Here, four different types of models were developed, including deep learning algorithms and regression models, and their performance in predicting BG levels was compared within 30 or 60 min forecast horizons using data from 12 individual T1D patients. The two deep learning models are Dilated Convolutional Neural Networks (DCNN) model and a Sequence to Sequence Long Short Term Memory (Seq-to-Seq LSTM) model. The two regression models are a Multiple Linear Regression (MLR) model and a Bidirectional Reservoir Computing (BRC) model.

This paper has two main contributions: i) Models are trained and tested on a large-scale, multiple features dataset consisting of 12 T1D data contributors. A detailed discussion of 30 and 60 min time horizon prediction results is provided. ii) To the best of our knowledge, the work presented here is the first to show that a model based on linear regression can outperform state of the art deep learning models. To this end, a novel method of constructing a multi-step linear regression prediction model is developed.

The remainder of the paper is organized as follows. Section 2 presents the current state-of-the-art of data-driven methods specific for the BG level prediction task. Section 3 describes the dataset used to evaluate the models. In Section 4, the four data-driven models are introduced. Section 5 presents the results followed by a discussion of the model behavior in Section 6. Section 7 provides conclusion remarks and suggests future research directions.

2. Related works

Many data-driven models have been tested for near-time BG level forecasts in the diabetes research community. In the category of regression models, the AR model proposed in [37] provides BG predictions as a linear combination of previously observed signals. A first-order AR model was proposed in [38] using time-varying parameters and was evaluated in 30 and 45 min prediction horizons.

Several researchers compared the AR model with other predictive models. The recursive AR with exogenous input (ARX) model [8] outperformed a vanilla LSTM, temporal convolution network, and other regression models for the prediction of BG levels from CGM data. However, in the ARIMA model [13], characterizing the movement of time series using an AR integrated (ARI) and moving average (MA) strategy resulted in worse performance than a random forest model. In addition, the linear model in [27] resulted in poorer performance in accuracy metrics compared to LSTM networks.

Recent work also included the combination of regression models and neural networks. An AR Time Delayed (ARTiDe) jumps neural network model [12] added the feedback connections from the output layer to the hidden layer and time delays to each connection on an AR model. In [9], the stacked regression contained a Multilayer Perceptron (MLP), an LSTM, and a Partial Least Squares Regression model (PLSR), and used predictions from three models to train a second-level model. A similar stack model [10] was tested on a multi-lag system. Finally, a Residual Compensation Network (RCN) combined with an online ARMA model (RCN-ARMA) [11] was used for simulating the nonlinear trends in BG dynamics.

However, the AR models typically use only CGM data to compute the prediction recursively, and the errors may accumulate in the prediction phase. The utilization of heterogeneous features should be considered in the regression models [39]. In addition, the regression models sometimes do not outperform the deep learning algorithms in predicting chaotic time series tasks.

Developing models that can utilize and accurately predict heterogeneous sources of time series data is still an active area of research. In the framework of deep learning approaches, there are a few related studies that address the use of LSTM layers for BG predictions to avoid the gradient vanishing problem of vanilla RNNs [40]. In [22], a BG prediction model was trained with only CGM data using multiple LSTM layers and a fully connected output layer. In [32], three LSTM networks trained with a transfer learning strategy were evaluated on the Clark error grid [41]. In [33], the past observations and estimated future values provided inputs for two separate LSTM networks, which were then combined using a fully connected layer. The multiple lag strategy was applied to multi-scale LSTM [31] so the network can more effectively model the dependence between features and CGM data. A four-layer bidirectional LSTM was introduced in [24] which contained a look back period of time. To improve prediction accuracy, the CGM was transferred to a one-hot encoder vector in the LSTM classification model in [25]. Additionally, the encoder and decoder strategy was first tested in [34] for BG predictions, where Fox et al. developed multi-output deep learning architectures by representing the inputs to a shared layer and then learning through the use of output networks. Several sequence to sequence neural networks were tested in [29] with LSTM or convolutional layers.

Selecting an artificial neural network and determining the model hyperparameters are crucial steps that can affect the evaluation performance, but can also be time-consuming. For a large neural network model, it can result in a slow training speed and a large combination of hyperparameters to tune. This study aims to develop computational efficient models that minimize the tuning effort and training time.

3. Data processing

Available data. The OhioT1DM Dataset [42] was initially provided to test the accuracy of models for the BG prediction task in the Blood Glucose Level Prediction (BGLP) Challenge in the third and fifth International Workshop on Knowledge Discovery in Healthcare Data. Each challenge released information from 6 data contributors, after the competition these data were publicly available. All 12 anonymous data contributors (labeled with numbers such as 540, 544, 563, etc.) have a total of around eight weeks of measurements from CGM, insulin therapy, physiological sensors, self-reported events for smartphone applications, with approximately 10 days of data that were held out for out-of-sample testing purposes; the date, month, and year were shifted randomly into the future time to anonymize the data.

The data includes 7 males and 5 females between ages 20 and 60. The constantly worn insulin pump (Medtronic 530G or 630G) measured the data contributors' BG levels every 5 min and the bolus and basal insulin dose injection amount; a smartphone recorded self-reported events including carbohydrate consumption, sleep quality, illness level, stress

level, hypo-event, work intensity, and exercise duration; and fitness bands (Basis Peak or Empatica Embrace) collected heart rate, room temperature, skin temperature, moving acceleration, step count, and Galvanic Skin Response (GSR). The CGM calibration is carried out every 6-8 h per day by self-measured fingerstick values.

Data time grid. The time grid is uniformly arranged with 5 min intervals between each time point for all features since the CGM data are collected every 5 min. The other features including carbohydrate intake, skin temperature, acceleration, GSR, fingerstick values, and bolus insulin dose are moved to the nearest time grid point when there exist misaligned measurements with the CGM signals. For categorical variables such as illness and hypo-events, event occurrences are represented by an integer value equal to 1 after moving to the corresponding time grid. The interval variables for which features are collected in a continuous period, such as basal insulin dose, exercise duration, and work intensity values, are filled in for the time range where data contributors had the activities.

Feature selection. The features with small incidence (the ratio of number of events to the total length of the dataset is $< 1\%$) were removed for the data contributor when training the predictive models. If a categorical feature is selected for training purposes, but completely missing in the testing set, then the feature will be added in the testing by setting all elements equal to 0. For example, if illnesses are reported in the training set but not in the testing set, we assume the contributor does not get ill in the testing period and set the corresponding categorical variable equal to 0. The four data-driven models are personalized for each individual, thus each data contributor may have a different number of features, the list of specific features used in each data contributor for all data-driven models is displayed in Table 1. For a data contributor, the same features were applied to every model. The common features are CGM, fingerstick values, basal and bolus insulin dose, carbohydrate consumption, GSR, and skin temperature.

Data imputation. Linear interpolation for continuous variables is used to connect the endpoints of intervals with missing values in the training data. In the test set, the CGM measurements with large missing intervals (over 6 h) are excluded, and forward filling is used to impute the rest of the missing data, i.e., by carrying forward the last observation with an existing value. For the categorical variables and interval variables, the missing values are replaced with 0 for both training and testing data.

Data scaling. The range of values varies extensively for each feature. Data were normalized by subtracting the mean and dividing by the standard deviation.

Table 1
Training features used by the 12 data contributors (dc) for the proposed models. The first column denotes the number of data contributors for which models were developed using the features in the third column. The second column denotes the number of features in the third column. The table is presented from the most features to the least features.

#dc	#features	Feature list
1	14	CGM, fingerstick, basal and bolus insulin dose, carbohydrate consumption, GSR, skin temperature, sleep quality, illness, work intensity, exercise duration, heart rate, room temperature, steps
3	13	CGM, fingerstick, basal and bolus insulin dose, carbohydrate consumption, GSR, skin temperature, sleep quality, work intensity, exercise duration, heart rate, room temperature, steps
1	13	CGM, fingerstick, basal and bolus insulin dose, carbohydrate consumption, GSR, skin temperature, sleep quality, work intensity, hypo-event, heart rate, room temperature, steps
1	13	CGM, fingerstick, basal and bolus insulin dose, carbohydrate consumption, GSR, skin temperature, sleep quality, illness, exercise duration, heart rate, room temperature, steps
2	9	CGM, fingerstick, basal and bolus insulin dose, carbohydrate consumption, GSR, skin temperature, acceleration
4	8	CGM, fingerstick, basal and bolus insulin dose, carbohydrate consumption, GSR, skin temperature

Data splitting. To observe the behavior and save the best performing model, validation data is necessary. The training set is further separated by using the last 20% of time points for validation. Thus, for the average training duration of 40 days, the last 8 days are used for validation. Table 2 shows the average length of time-series data and the missing percentage of continuous variables for training, validation and testing sets over the 12 data contributors.

Data expansion. To increase the number of training samples, the multi-horizon method is introduced. Previous work [43] suggested that the multi-horizon strategy is accurate for short time series forecasts. It can also control overfitting and is able to recover potential covariate relationships. The training data of each contributor is truncated in a sliding window, moving 1 time-step forward with a fixed window size.

4. Methods

The four data-driven models are presented, including two regression models and two deep neural network models.

4.1. Multiple linear regression model

Due to the structure of linear regression, fitting three-dimensional data adds difficulty in solving the inverse problem. The data expansion is omitted, and the validation set is unnecessary since there are no hyperparameters to tune. To train a model offline and compare it with the other three models described in the following section, we used the combination of the multi-step-ahead strategy to generate multi-horizon predictions as shown in Fig. 1.

The Multiple Linear Regression (MLR) model is constructed by combining multiple linear regression models. In this approach, k separate linear regression models, denoted by L_i, are used for k time step ahead predictions. Each linear regression model L_i is trained to find the relationship between $X_{train}(q_1(t), \ldots, q_m(t))$ and $y_{train}(t+i)$, where $i = 1, 2, \ldots, k$ for $k = 6$ or 12 represents 30 or 60 min prediction, respectively. $X(\cdot)$ is the matrix containing m number of feature values $q(t)$ at time t, where for a specific data contributor, the value m can be found in Table 1, and $y(t)$ represents the CGM value at time t.

After training and obtaining the coefficients and intercepts for k linear regression models (i.e., $L_1, L_2, \ldots, L_k$), predictions are computed by using the k linear models from the testing data. Specifically, data measurements one time step before the first test point are used, denoted by $X_{test}(q_1(t_0), \ldots, q_m(t_0))$. The trained coefficients and intercepts of the k linear models are used to predict the BG levels for the next k time steps in the forecast horizon, i.e., $[\hat{y}_{test}(t_1), \hat{y}_{test}(t_2), \ldots, \hat{y}_{test}(t_k)]$. After that, the next row of the testing matrix is used calculate predictions from t_2 to t_{k+1}. This method continues to the end of the testing matrix until all future predictions in k time step horizons are obtained. In this way, if a T1D patient had data measurements up to 10 a.m., then the model can give a warning if any hypoglycemia would happen in between 10 a.m. to 10:30 a.m. (using 6 linear models) or between 10 a.m. to 11:00 a.m. (using 12 linear models). The performance is evaluated by comparing each prediction vector with the corresponding labels in that time range. The values used in the prediction plot are found by averaging the multi-horizon forecasts.

4.2. Bidirectional reservoir computing model

Reservoir computing has much success in accurately predicting multivariate time-series data [44]. The core of reservoir computing is to build a network to process the measurements in a system-independent fashion. The Bidirectional Reservoir Computing (BRC) model is constructed from [45], which introduces the encoder-decoder framework that transforms the inputs by tuning model hyperparameters. The BRC model is constructed from three main components: a linear input layer, a bidirectional recurrent nonlinear reservoir network, and a linear output layer, as shown in Fig. 2.

Table 2
Average length and average percentage of the missing data for continuous variables for training, validation and testing sets over 12 data contributors when the features are available. CGM represents the continuous glucose monitoring, HR stands for heart rate, GSR is galvanic skin response, and T_{skin}, T_{room} stand for the skin temperature and room temperature, respectively. For continuous variables in the training and validation sets, the imputation method is linear interpolation. In the testing set, the forward filling method is used for imputing missing data and the large missing intervals of CGM are removed.

	Average	Average percentage of missing data				
	Length	CGM	Acceleration	HR	GSR, T_{skin}	T_{room}
Train	10061.83	12.13%	36.48%	4.17%	20.51%	4.53%
Validation	2515.17	11.67%	44.85%	7.70%	26.47%	8.08%
Test	2896.83	8.63%	30.78%	6.75%	18.94%	7.09%

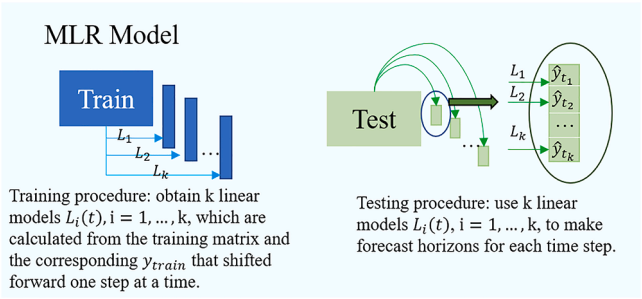


Fig. 1. Schematic diagram of the MLR model using multi-horizon prediction, where $k = 6, 12$ represents the 30 or 60 min, respectively.

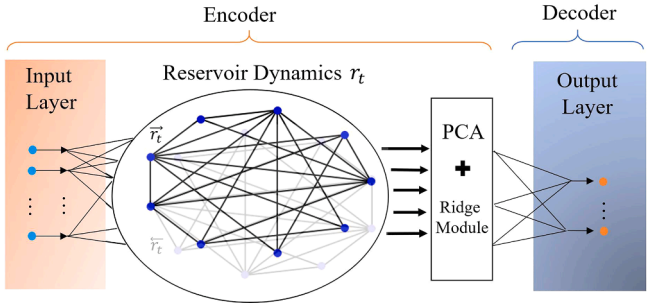


Fig. 2. Schematic diagram of the BRC model, r_t is the reservoir state at time t.

In the encoder part of the model, the inputs contain the list of features in Table 1 with a fixed window size for a data contributor and trained through bidirectional reservoir networks, which concatenates the bidirectional state vectors that were trained through the same reservoir networks for the reverse direction. The forward direction state vector $\overrightarrow{r_t}$ is updated by the following equation:

$$\overrightarrow{r_t} = \left(1-\gamma\right)\overrightarrow{r}_{t-1} + \tanh\left(W_{res}\cdot\overrightarrow{r}_{t-1} + W_{in}\cdot x_t + b_{res}\right), \quad (1)$$

where γ is a scalar that determines what proportion we want to update by the previous state, x_t represents the input information at time t, W_{res} are the internal weights that are initialed by a sparse matrix with uniformly distributed values in the range $[-0.5, 0.5]$. These uniformly distributed values are divided by a scalar computed from the maximum eigenvalues of the internal weight matrix over a spectral radius. The input weights W_{in} are initialized by a binomial distribution multiplied by input scaling. Lastly, b_{res} represents biases which are the product of a randomized noise matrix and a noise level.

The backward direction state vector $\overleftarrow{r_t}$ is computed by flipping the second dimension of x_t in Eq. (1). The updated state vector r_t uses the bidirectional reservoir state, which enables the model to generate improved multivariate time series predictions. Next, the dimensionality is significantly reduced by Principal Component Analysis (PCA). Then, the reduced network is parameterized with ridge regression to get the representation of the reservoir model.

For the decoder part of the model, ridge regression reads the output from the encoder and transfers the information back to obtain multiple prediction horizons. By setting a large ridge regression parameter α in Eq. (2), the weights are balanced to avoid overfitting.

Ridge regression computes the weights w by finding the minimum solution for the input X and output y using the following cost function:

$$\min_{w}\left(\left|\left|y - Xw\right|\right|_2^2 + \alpha\left|\left|w\right|\right|_2^2\right), \quad (2)$$

where X and y correspond with the reshaped matrices after the PCA dimensionality reduction in the representation module. For ridge regression in the decoder, X are the weights obtained from the representation module, and y is the CGM signals shifted 30 or 60 min into the future.

The computational training and testing times depend on the size of the reservoir and input time steps. The validation data is used to tune the number of input time steps and the hyperparameters (i.e., γ, spectral radius, input scaling, etc.) for the reservoir dynamics to obtain both the efficiency and accuracy.

4.3. Dilated convolutional neural network model

WaveNet [46], a deep neural network that has been used to accurately generate audio waveforms, relies on a type of neural network layer called a dilated convolutional layer. This dilated convolution layer is used here to build a Dilated Convolutional Neural Network (DCNN) model. The model size is largely reduced and there are fewer hyperparameters to tune. The DCNN model is built from multiple DCNN layers and two time-distributed dense layers.

DCNN layer. The DCNN model uses the strategy of combining dilated convolutions and causal padding, where the dilated convolutions let the dilation rate increase multiplicatively at each layer to achieve an exponential relationship between receptive field size and the layer number, and the casual padding is used to avoid using future values.

Time-distributed dense layer. A dense layer calculates the output, o, using an activation function f involving weights W and biases b, denoted by $o = f(Wx + b)$, where x is the input. Since separating each time step for time series is necessary, a time-distributed layer is applied to each dense layer for each time step.

As depicted in Fig. 3, the multiple DCNN layers (kernel size $= 2$)

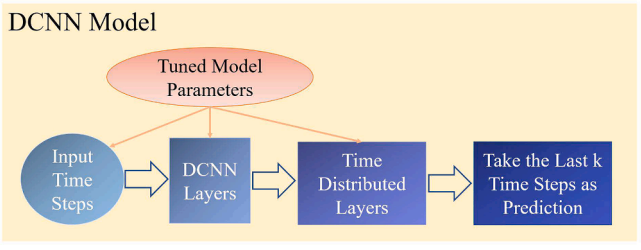


Fig. 3. Schematic diagram of the DCNN model.

with causal padding are used and L_2 regularization is applied to train the weights in each layer. In the first time-distributed layer, the ReLu activation function is used with L_2 regularization. Then, a dropout layer is used to scale down the output, which is then input to the next time distributed layer. The second time distributed layer is used to compute the final output using a linear function with L_2 regularization. The last 6 or 12 time steps are selected for computing predictions over 30 and 60 min time horizons, respectively. The input time series for one data contributor has the features presented in Table 1, the corresponding target time series are the CGM signals shifted either 30 or 60 min into the future.

The model hyperparameters are tuned by using grid search to determine the best combination of parameters of each data contributor, including input time steps (selected from [72, 144, 288, 576]), the number of DCNN layers (selected from [2–4]), and the number of neurons in the hidden layers (selected from [16,32]). The loss function used in both training and validation is defined as follows.

Loss function. The Weighted Mean Squared Error (WMSE) is calculated by the following formula,

$$WMSE = \sum_{i=1}^{N}\left((\hat{y}_i - y_i)^2 \times \frac{y_i}{M}\right) \tag{3}$$

where y_i presents the true value and $\hat{y}_i$ stands for the predicted value at time $i \in [1, N]$. M is a fixed scalar, usually selected slightly larger than the maximum of the data values. Compared to the mean squared error, WMSE focuses on amplifying the error in mispredicting hyperglycemia, which is usually underestimated in the training phase.

4.4. Sequence to sequence long short term memory model

Sequence to sequence (Seq-to-Seq) models [47] use an encoder-decoder framework that can map sequences of different lengths from input to output.

An LSTM layer [40] is a recurrent neural network layer with a chain of repeating modules, each module contains three interacting gates: a forget gate, input gate, and output gate. The network in the LSTM layer can be explained as follows. The information from the previous state is first computed by the sigmoid function to determine which part to forget (forget state), then a second sigmoid function determines the information to carry on (input gate), and lastly, the output information obtained by the output gate from the third sigmoid function. The internal memory combines the input state and drops the forgotten information from the old cell, and the hidden state is updated through an activation function that is applied to a combined vector consisting of the internal memory and the information from the output gate.

The Seq-to-Seq LSTM model applies the encoder-decoder strategy using LSTM layers. In the encoder LSTM, the current hidden state is propagated from the previous hidden state and input vector, where the input vector is taken from each LSTM unit. The encoder vector encapsulates the information in the encoder phase and provides it to the decoder. Finally, the decoder LSTM includes a stack of several recurrent units where each unit provides the output at time t using the linear activation function as shown in Fig. 4.

A window of 6 h with the training features listed in Table 1 is provided for the input series, and the decoder input series is the CGM signals shifted forward by one time step with zero padding used in the first time step. For time t, the decoder input series which works as a mock target contains $[0, Y(t), Y(t+1), \ldots, Y(t+k-1)]$, and the target decoder output is $[Y(t), Y(t+1), Y(t+2), \ldots, Y(t+k)]$, for the time-series $Y(t)$ representing the future 30 or 60 min of CGM signals at time t from the training and validation. In the predicting phase, the testing time series was fed into the encoder model with one time step at a time. Then the encoder model outputs the state vector which becomes the input state for the decoder model. The decoder model iteratively computes each element in the prediction vector by updating from the previous predicted value and output states. The model uses an LSTM layer with 128 recurrent units for both the encoder and decoder and summarizes the output through a dense layer. The behavior of the validation set is monitored by the mean squared error and the training is stopped if the loss function is stable after 8 epochs. The best performing model is applied to the time series data in the test set to obtain predictions.

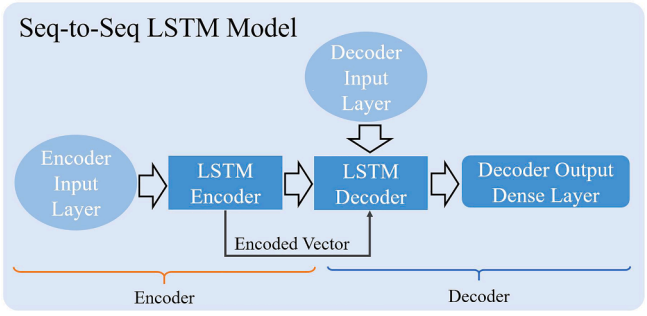


Fig. 4. Schematic diagram of the Seq-to-Seq LSTM model.

4.5. Evaluation metrics

The predictive accuracy for BG levels in training, validation, and test data are evaluated by Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE) defined below. During the evaluation, the imputed value of CGM in training, validation and test data are excluded. In the test data, the first hour is excluded for utilizing the full length of training and validation data in the training procedure.

$$RMSE = \sqrt{1/N\sum_{i=1}^{N}(\hat{y}_i - y_i)^2}, \tag{4}$$

$$MAE = 1/N\sum_{i=1}^{N}|\hat{y}_i - y_i|, \tag{5}$$

where the $\hat{y}_i$, y_i are the predicted value and true value for time index i, respectively. N is the total number of data points evaluated. We note that the RMSE is more sensitive to outliers than the MAE.

5. Results

The evaluation tables for 30 and 60 min forecast horizons are shown in Tables 3 and 4 for four data-driven models. All results are averaged over all data contributors. We observe that the Seq-to-Seq LSTM model performed the best in 30 min ahead predictions and the MLR model performed the best in 60 min ahead predictions. An example of a prediction plot for each model, using the testing set from one data contributor, is depicted in Figs. 5 and 6.

The two evaluation metrics (RMSE and MAE) show a large difference among the four data-driven models, likely reflecting the fact that CGM signals have large time-varying fluctuations. We found that the four

Table 3
Table of average RMSE and MAE evaluations within a 30 min forecast horizon for 4 proposed data-driven models. The best testing performance is labeled in bold.

	RMSE (mg/dl)			MAE (mg/dl)		
Model	Train	Validation	Test	Train	Validation	Test
MLR	14.16	-	18.39	9.87	-	11.76
BRC	10.82	19.29	24.49	7.74	13.50	17.43
DCNN	21.41	23.55	26.96	15.60	16.76	18.25
Seq-to-Seq LSTM	10.02	13.83	**17.52**	7.11	9.26	**10.91**

Table 4
Table of average RMSE and MAE evaluations within a 60 min forecast horizon for 4 proposed data-driven models. The best testing performance is labeled in bold.

	RMSE (mg/dl)			MAE (mg/dl)		
Model	Train	Validation	Test	Train	Validation	Test
MLR	21.39	–	**24.58**	15.56	–	**17.42**
BRC	13.95	24.93	28.79	10.12	18.10	20.75
DCNN	35.84	36.98	38.97	27.37	27.96	29.38
Seq-to-Seq LSTM	18.22	22.40	26.44	13.96	16.54	18.43

data-driven models had difficulty capturing the extreme values in the training, validation, and testing sets. Additionally, we observed that the 60 min time horizon predictions were less likely to capture the extreme values compared to the 30 min predictions, especially when using the deep learning models. The DCNN model is trained on a weighted loss function, which may lead to more fluctuations in the predictions when the CGM signal is smooth.

Two encoder-decoder models were trained, namely the BRC and Seq-to-Seq LSTM models. Compared with other models in the same category (regression or deep learning), the model with encoder-decoder framework has more variance than the input CGM features, while the MLR and DCNN mainly follow the original CGM inputs with variations due to the model structure. With regards to the ability to accurately predict missing data, we found that the encoder-decoder models produced predictions that were similar to the fluctuating trends around the missing regions, whereas the other two models stayed flat especially in 60 min predictions (see Figs. 5 and 6 with x-axis labeled with the red triangles around day Jan. 21).

Comparing computational efficiency, the DCNN, Seq-to-Seq LSTM, BRC, and MLR models took an average of 30 min, 10 min, 20 s, and 3 s

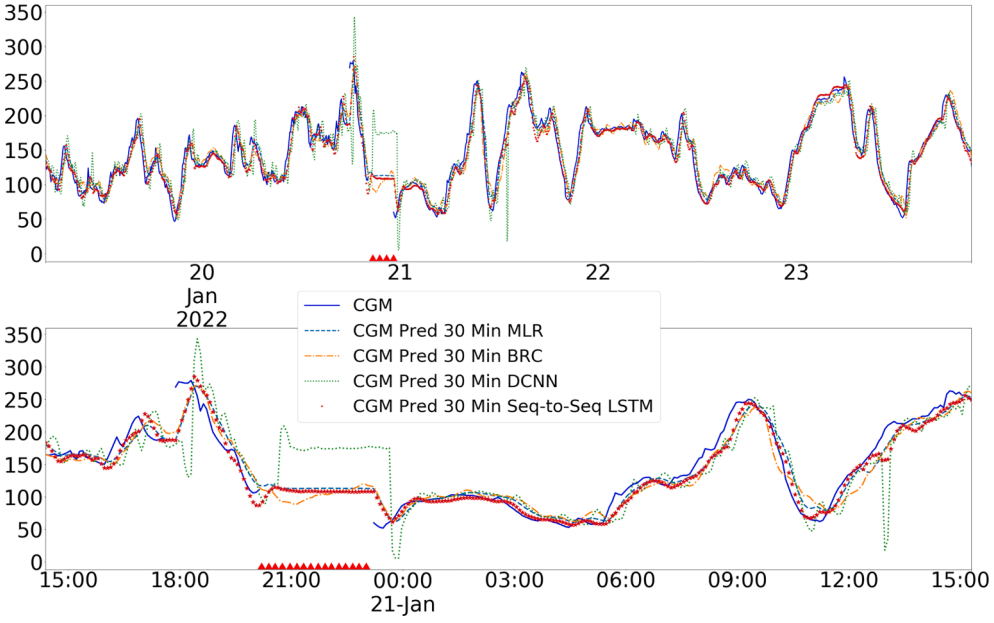


Fig. 5. Comparison of forecasting plots for the four proposed models using the testing set of data contributor 591 in 30 min multi-horizon predictions. The blue curve represents CGM signals, cyan dashed line represents CGM prediction for MLR model, orange dash-dot line stands for BRC model, green dotted line is for DCNN model, and red star is the Seq-to-Seq LSTM. The red triangles in the x-axis denote the date around Jan. 21 where data for CGM is missing. The bottom subplot is a zoom that displays the time around the missing data interval from Jan. 21.

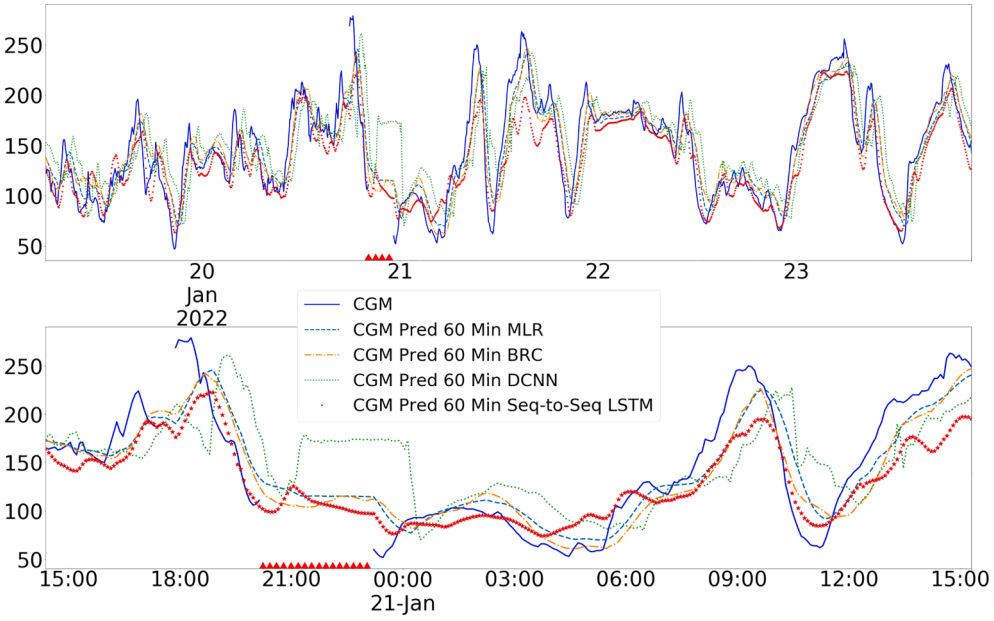


Fig. 6. Comparison of forecasting plots for the four proposed models using the testing set of data contributor 591 in 60 min multi-horizon predictions. The blue curve represents CGM signals, cyan dashed line represents CGM prediction for MLR model, orange dash-dot line stands for BRC model, green dotted line is for DCNN model, and red star is the Seq-to-Seq LSTM. The red triangles in the x-axis denote the date around Jan. 21 where data for CGM is missing. The bottom subplot is a zoom that displays the time around the missing data interval from Jan. 21.

Table 5
Comparison of results from the Third International Workshop on Knowledge Discovery in Healthcare Data. RMSE (mg/dl) is evaluated within 30 and 60 min prediction horizons. If a study did not conduct the experiment for 60 min, then it is labeled with –. The methods and corresponding authors are presented in the order from the worst to best RMSEs for 30 min. We list the models considered in this work as "This study" for comparison. Bold font indicates instances in which our models showed a lower RMSE than previous models.

Study	Method	RMSE 30 min	RMSE 60 min
Zhu et al. [21]	Causal CNN	27.73	–
Contreras et al. [15]	GE	21.19	31.34
Midroni et al. [14]	XGBoost	20.38	–
Martinsson et al. [22]	LSTM	20.1	33.2
Xie, J. and Wang, Q. [8]	ARX	19.59	–
Bertachi et al. [35]	Hybrid model	19.33	31.72
Chen et al. [23]	DRNN	19.04	–
This study	MLR	**15.46**	**21.59**
	BRC	20.92	**24.67**
	DCNN	24.98	36.39
	Seq-to-Seq LSTM	**14.87**	**23.45**

for training each data contributor, respectively. These results strongly suggest that the MLR model is significantly faster to train than deep learning models for the complex multivariate time series data considered in this work.

The differences in performance in the training, validation, and testing sets for the four data-driven models may have resulted from overfitting. In practice, the training performance should always be better than performance on the validation or testing sets. Due to the different imputation methods in the validation and testing sets, the evaluation results indicate that the imputation method may affect the prediction performance according to the model type. For the BRC model, the training error is significantly lower than validation and testing error. Even with setting the ridge regulation parameter α in Eq. (2) to be a large number (e.g., 10^2), the results are not optimal. However, the training performance of the BRC model is similar to the Seq-to-Seq LSTM model for 30 min predictions and outperforms all other models in 60 min predictions, which demonstrates the potential of the BRC model in time-series forecasting tasks.

6. Discussion

We compared the accuracy of our models with previous models that were developed for the BGLP challenge on the OhioT1DM dataset. Table 5 shows results comparing our models to models in the first BGLP challenge, which used data from the first six data contributors in the OhioT1DM dataset. We note that the performance of all models is reported only with the RMSE, and mostly only for 30 min prediction horizons. Table 6 shows results compared to models in the second BGLP challenge, which used data from the remaining six data contributors in the OhioT1DM dataset. The table reports the RMSEs and MAEs for 30 and 60 min prediction horizons. Overall, the results in Tables 5 and 6 indicate that both our models and previously developed models show large differences in the RMSE and MAE metrics.

The third column in Table 5 indicates that our MLR and Seq-to-Seq LSTM models outperformed previously published models, and the fourth column shows that three of our models were more accurate at 60 min predictions when the dataset does not have large missing intervals, as we found only two contributors had large missing intervals in the testing data among the 6 evaluated contributors. In Table 6, our MLR and Seq-to-Seq LSTM models outperformed previously published models for 60 min predictions using the RMSE evaluation metric, and obtained lower MAEs compare to most models except NN-EIM [20]. Among the 6 data contributors, five data contributors had large missing intervals in the testing set. The failure to obtain better results for 30 min predictions may be due to the data processing method we used for the testing data.

In the testing set, we skipped the missing intervals that are above 6 continuous hours of missing CGM signals, and imputed the rest of data by the forward filling. However, the imputation method will largely

Table 6
Comparison of results from the Fifth International Workshop on Knowledge Discovery in Healthcare Data. RMSE (mg/dl) and MAE (mg/dl) are evaluated within 30 and 60 min prediction horizons. The methods and corresponding authors are presented in the order from the worst to best RMSEs for 30 min. We list the models considered in this work as "This study" for comparison. Bold font indicates instances in which our models showed a lower RMSE than previous models.

Study	Method	RMSE 30 min	MAE 30 min	RMSE 60 min	MAE 60 min
Bhimireddy et al. [29]	Seq-to-Seq BiLSTM	21.8	15.0	35.0	25.0
Cappon et al. [24]	BLSTM	20.2	14.74	34.19	25.98
Joedicke et al.[16]	GP	20.13	14.05	32.18	23.58
Ma et al. [11]	RCN-ARMA	20.03	14.52	34.89	26.41
Daniels et al. [28]	MTCRNN	19.79	13.62	33.73	24.54
Mayo et al. [25]	LSTM-NN	19.4	13.9	33.4	25.0
Sun et al. [18]	LV-based model	19.37	13.76	32.59	24.64
Hameed et al. [26]	Vanilla RNN	19.21	13.07	31.77	23.09
Yang et al. [31]	MS-LSTM	19.05	13.50	32.03	23.83
Khadem et al. [10]	Multi-lag stacking	19.01	13.73	33.37	24.98
Nemat et al. [9]	Stacked regression	18.99	13.73	33.39	25.04
Pavan et al. [20]	NN-EIM	18.63	10.08	32.27	17.69
Zhu et. al. [19]	GAN	18.34	13.37	32.21	24.20
Bevan, R. and Coenen, F. [27]	LSTM	18.23	14.37	31.1	25.75
Rubin-Falcone et al. [17]	N-BEATS	18.22	12.83	31.66	23.60
Freiburghaus et al. [30]	CRNN	17.45	11.22	33.67	23.25
This study	MLR	21.32	13.13	**27.56**	19.10
	BRC	28.06	19.15	32.92	23.80
	DCNN	28.93	19.41	41.24	31.28
	Seq-to-Seq LSTM	20.17	11.79	**29.44**	19.79

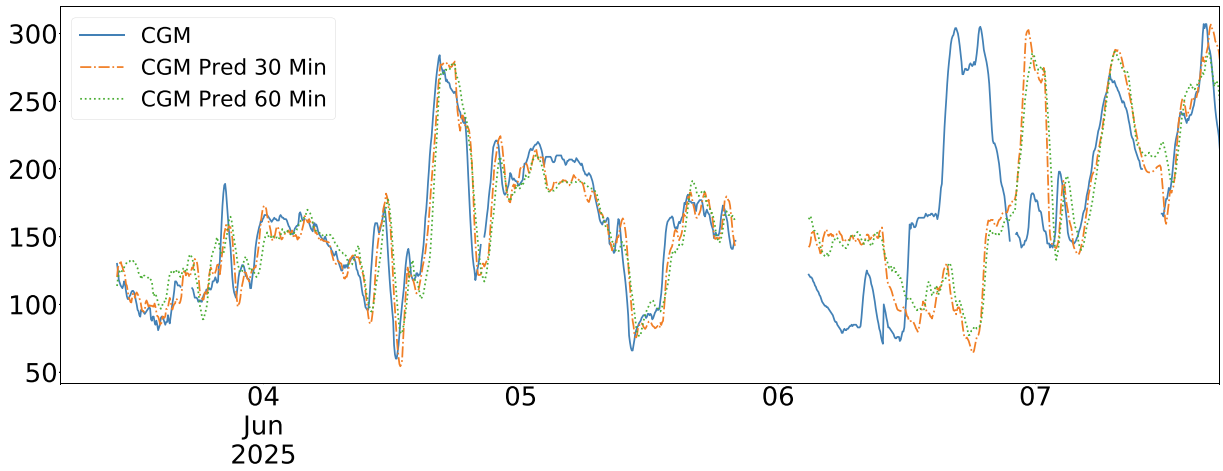


Fig. 7. Forecast plots for the BRC model using the testing set of data contributor 552. The cyan curve represents CGM signals, orange dashed line is BG level prediction for a 30 min time horizon, and green dotted line is the prediction for a 60 min time horizon. Note that the date and time were randomly shifted to future date and time in the original OhioT1DM dataset. The missing interval around June 6th is removed when the prediction is calculated.

affect prediction performance. For example, in Fig. 7, an interesting behavior was observed by the BRC model for one data contributor where the values around June 6th are largely missing. After the missing data, the BRC model starts to predict CGM levels incorrectly. After approximately one day, e.g., at approximately June 7th, the BRC model begins producing more accurate predictions.

The example in Fig. 7 reveals two issues: first, we should not remove the large missing intervals. Second, the evaluation by RMSE, MAE, or other metrics may not detect the poor prediction behavior that occurs just after the missing data intervals. Additionally, the low RMSE may be due to the shifted behavior of CGM signals when fewer data points with fluctuations are presented [21]. Together, these results suggest that the evaluation metrics should always be combined with the visualization of prediction plots for accurate interpretation.

Among the 12 individual data contributors, we found that the performance was highly variable. The data contributors with lower fluctuations in CGM levels and fewer missing values tended to result in better performance in evaluation metrics for all models. For data contributors with a highly fluctuating pattern in CGM levels (perhaps due to an unstable lifestyle) that contained a large amount of missing values, the MLR model outperforms other models. Thus, for large-scale time-series data over a population of individuals, these results suggest it is necessary to test multiple models and that performance of a specific type of model may depend on data heterogeneities in data omission and levels of fluctuation.

Notably, we found that better prediction performance was achieved with regression models for short time series forecasts. Overall, our novel MLR model achieves a good performance in predicting a 30 and 60 min horizon from a single time point. The model utilizes multiple linear regressions to obtain predictions k time steps in the future and works better than AR models in multi-horizon forecasts, since the latter may accumulate errors during calculations. The BRC model could more accurately capture the possible trend of future CGM levels during the forward filling intervals, and the prediction difference between 30 and 60 min is relatively small. Lu et al. [44] note that the performance of reservoir computing models may depend on initial parameters and can sometimes be very effective. By carefully tuning the model parameters and using a better imputation method, overfitting can be avoided.

7. Conclusion

In this work, four data-driven models using regression or deep learning were introduced to obtain multi-horizon predictions in patients with T1D. Among these models, the Seq-to-Seq LSTM model performs the best at predicting BG levels within a 30 min horizon and the MLR model is the best at predicting BG levels within a 60 min horizon. These models could be used to develop an artificial pancreas that helps control BG concentrations in a closed-loop system. We presented an analysis based on the visualization of BG prediction plots, and found that the Seq-to-Seq LSTM and regression models have the potential to make highly accurate predictions.

In future research, the prediction of hypo/hyper-glycemia classification could also be considered and compared with the performance of regression models [7]. The proposed data-driven models can also be extended to a longer prediction horizon and explored within the framework of hybrid models, i.e., where physiological models [4–6] are combined with the best performing data-driven models. The BG level forecasts produced by the computationally efficient models considered in this work may be added in a visualization tool on the used for a closed-loop insulin delivery system for T1D patients [48], thereby improving the system interface to encourage healthy activity and learning.

Funding source

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

CRediT authorship contribution statement

Meng Zhang: Methodology, Data curation, Software, Formal analysis, Writing - original draft, Writing - review & editing. **Kevin B. Flores:** Methodology, Conceptualization, Writing - review & editing. **Hien T. Tran:** Methodology, Conceptualization, Supervision, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

[1] M. Mobasseri, M. Shirmohammadi, T. Amiri, N. Vahed, H. HosseiniFard, M. Ghojazadeh, Prevalence and incidence of type 1 diabetes in the world: a systematic review and meta-analysis, Health Promotion Perspectives 10 (2) (2020) 98–115. [Online]. Available: doi: 10.34172/hpp.2020.18.

[2] E. Georga, D. Fotiadis, S. Tigas, Personalized Predictive Modeling in Type 1 Diabetes, Academic Press, Ioannina, Greece, 2018. [Online]. Available: URL:

https://www.sciencedirect.com/book/9780128048313/personalized-predictive-modeling-in-type-1-diabetes.
[3] M. Atkinson, G. Eisenbarth, A. Michels, Type 1 diabetes, The Lancet 383 (9911) (2014) 69–82. [Online]. Available: doi: 10.1016/S0140-6736(13)60591-7.
[4] R. Bergman, L. Phillips, C. Cobelli, Physiologic evaluation of factors controlling glucose tolerance in man: measurement of insulin sensitivity and beta-cell glucose sensitivity from the response to intravenous glucose, Journal of Clinical Investigation 68 (6) (1981) 1456–1467. [Online]. Available: doi: 10.1172/jci110398.
[5] C. Man, R. Rizza, C. Cobelli, Meal simulation model of the glucose-insulin system, IEEE Transactions on Biomedical Engineering 54 (10) (2007). [Online]. Available: URL: https://ieeexplore.ieee.org/document/4303268.
[6] C. Man, M. Breton, C. Cobelli, Physical activity into the meal glucose-insulin model of type 1 diabetes: In silico studies, Journal of Diabetes Science and Technology 3 (1) (2009). [Online]. Available: URL: https://journals.sagepub.com/doi/10.1177/193229680900300107.
[7] A.Z. Woldaregay, E. Årsand, T. Botsis, D. Albers, L. Mamykina, G. Hartvigsen, Data-Driven Blood Glucose Pattern Classification and Anomalies Detection: Machine-Learning Applications in Type 1 Diabetes, J Med Internet Res 21 (5) (2019) e11030. [Online]. Available: doi: 10.2196/11030.
[8] J. Xie and Q. Wang, Benchmark machine learning approaches with classical time series approaches on the blood glucose level prediction challenge, in: CEUR Workshop Proceedings, vol. 2148, 2018, pp. 97–102. [Online]. Available: URL: http://ceur-ws.org/Vol-2148/paper16.pdf.
[9] H. Nemat, H. Khadem, J. Elliott, M. Benaissa, Data fusion of activity and cgm for predicting blood glucose levels, in: CEUR Workshop Proceedings, vol. 2675, 2020, pp. 120–124. [Online]. Available: URL: http://ceur-ws.org/Vol-2675/paper21.pdf.
[10] H. Khadem, H. Nemat, J. Elliott, M. Benaissa, Multi-lag stacking for blood glucose level prediction, in: CEUR Workshop Proceedings, vol. 2675, 2020, pp. 146–150. [Online]. Available: URL: http://ceur-ws.org/Vol-2675/paper26.pdf.
[11] N. Ma, Y. Zhao, S. Wen, T. Yang, R. Wu, R. Tao, X. Yu, H. Li, Online blood glucose prediction using autoregressive moving average model with residual compensation network, CEUR Workshop Proceedings vol. 2675, 2020, pp. 151–155. [Online]. Available: URL: http://ceur-ws.org/Vol-2675/paper27.pdf.
[12] F. D'Antoni, M. Merone, V. Piemonte, G. Iannello, P. Soda, Auto-regressive time delayed jump neural network for blood glucose levels forecasting, Knowledge-Based Systems 203 (2020) 106134. [Online]. Available: URL: https://www.sciencedirect.com/science/article/pii/S0950705120303890.
[13] I. Rodríguez-Rodríguez, I. Chatzigiannakis, J. Rodríguez, M. Maranghi, M. Gentili, M. Zamora-Izquierdo, Utility of Big Data in Predicting Short-Term Blood Glucose Levels in Type 1 Diabetes Mellitus Through Machine Learning Techniques, Sensors (Basel) 19 (2019). [Online]. Available: URL: https://www.mdpi.com/1424-8220/19/20/4482.
[14] C. Midroni, P. Leimbigler, G. Baruah, Predicting glycemia in type 1 diabetes patients: Experiments with xg-boost, in: CEUR Workshop Proceedings, vol. 2148, 2018, pp. 79–84, 2018. [Online]. Available: URL: http://ceur-ws.org/Vol-2148/paper13.pdf.
[15] I. Contreras, A. Bertachi, L. Biagi, S. Oviedo, J. Vehi, Using grammatical evolution to generate short-term blood glucose prediction models, in: CEUR Workshop Proceedings, vol. 2148, 2018, pp. 91–96. [Online]. Available: URL: http://ceur-ws.org/Vol-2148/paper15.pdf.
[16] D. Joedicke, O. Garnica, G. Kronberger, J.M. Colmenar, S. Winkler, J.M. Velasco, S. Contador, J.I. Hidalgo, Analysis of the performance of genetic programming on the blood glucose level prediction challenge 2020, in: CEUR Workshop Proceedings, vol. 2675, 2020, pp. 141–145. [Online]. Available: URL: http://ceur-ws.org/Vol-2675/paper25.pdf.
[17] H. Rubin-Falcone, I. Fox, J. Wiens, Deep residual time-series forecasting: Application to blood glucose prediction, in: CEUR Workshop Proceedings, vol. 2675, 2020, pp. 105–109. [Online]. Available: URL: http://ceur-ws.org/Vol-2675/paper18.pdf.
[18] X. Sun, M. Rashid, M. Sevil, N. Hobbs, R. Brandt, M.R. Askari, A. Shahidehpour, A. Cinar, Prediction of blood glucose levels for people with type 1 diabetes using latent-variable-based model, in: CEUR Workshop Proceedings, vol. 2675, 2020, pp. 115–119. [Online]. Available: URL: http://ceur-ws.org/Vol-2675/paper20.pdf.
[19] T. Zhu, X. Yao, K. Li, P. Herrero, P. Georgiou, Blood glucose prediction for type 1 diabetes using generative adversarial networks, in: CEUR Workshop Proceedings, vol. 2675, 2020, pp. 90–94. [Online]. Available: URL: http://ceur-ws.org/Vol-2675/paper15.pdf.
[20] J. Pavan, F. Prendin, L. Meneghetti, G. Cappon, G. Sparacino, A. Facchinetti, S.D. Favero, Personalized machine learning algorithm based on shallow network and error imputation module for an improved blood glucose prediction, in: CEUR Workshop Proceedings, vol. 2675, 2020, pp. 95–99. [Online]. Available: URL: http://ceur-ws.org/Vol-2675/paper16.pdf.
[21] T. Zhu, K. Li, P. Herrero, J. Chen, P. Georgiou, A deep learning algorithm for personalized blood glucose prediction, in: CEUR Workshop Proceedings, vol. 2148, 2018, pp. 74–78. [Online]. Available: URL: http://ceur-ws.org/Vol-2148/paper12.pdf.
[22] J. Martinsson, A. Schliep, B. Eliasson, C. Meijner, S. Persson, O. Mogren, Automatic blood glucose prediction with confidence using recurrent neural networks, in: CEUR Workshop Proceedings, vol. 2148, 2018, pp. 64–68. [Online]. Available: URL: http://ceur-ws.org/Vol-2148/paper10.pdf.
[23] J. Chen, K. Li, P. Herrero, T. Zhu, P. Georgiou, Dilated recurrent neural network for short-time prediction of glucose concentration, in: CEUR Workshop Proceedings, vol. 2148, 2018, pp. 69–73. [Online]. Available: URL: http://ceur-ws.org/Vol-2148/paper11.pdf.
[24] G. Cappon, L. Meneghetti, F. Prendin, J. Pavan, G. Sparacino, S.D. Favero, A. Facchinetti, A personalized and interpretable deep learning based approach to predict blood glucose concentration in type 1 diabetes, in: CEUR Workshop Proceedings, vol. 2675, 2020, pp. 75–79. [Online]. Available: URL: http://ceur-ws.org/Vol-2675/paper12.pdf.
[25] M. Mayo, T. Koutny, Neural multi-class classification approach to blood glucose level forecasting with prediction uncertainty visualisation, in: CEUR Workshop Proceedings, vol. 2675, 2020, pp. 80–84. [Online]. Available: URL: http://ceur-ws.org/Vol-2675/paper13.pdf.
[26] H. Hameed, S. Kleinberg, Investigating potentials and pitfalls of knowledge distillation across datasets for blood glucose forecasting, in: CEUR Workshop Proceedings vol. 2675, 2020, pp. 85–89. [Online]. Available: URL: http://ceur-ws.org/Vol-2675/paper14.pdf.
[27] R. Bevan, F. Coenen, Experiments in non-personalized future blood glucose level prediction, in: CEUR Workshop Proceedings vol. 2675, 2020, pp. 100–104. [Online]. Available: URL: http://ceur-ws.org/Vol-2675/paper17.pdf.
[28] J. Daniels, P. Herrero, P. Georgiou, Personalised glucose prediction via deep multitask networks, in: CEUR Workshop Proceedings vol. 2675, 2020, pp. 110–114. [Online]. Available: URL: http://ceur-ws.org/Vol-2675/paper19.pdf.
[29] A. Bhimireddy, P. Sinha, B. Oluwalade, J.W. Gichoya, S. Purkayastha, Blood glucose level prediction as time-series modeling using sequence-to-sequence neural networks, in: CEUR Workshop Proceedings vol. 2675, 2020, pp. 125–130. [Online]. Available: URL: http://ceur-ws.org/Vol-2675/paper22.pdf.
[30] J. Freiburghaus, A. Rizzotti-Kaddouri, F. Albertetti, A deep learning approach for blood glucose prediction of type 1 diabetes, in: CEUR Workshop Proceedings vol. 2675, 2020, pp. 131–135. [Online]. Available: URL: http://ceur-ws.org/Vol-2675/paper23.pdf.
[31] T. Yang, R. Wu, R. Tao, S. Wen, N. Ma, Y. Zhao, X. Yu, H. Li, Multi-scale long short-term memory network with multi-lag structure for blood glucose prediction, in: CEUR Workshop Proceedings, vol. 2675, 2020, pp. 136–140. [Online]. Available: URL: http://ceur-ws.org/Vol-2675/paper24.pdf.
[32] S. Faruqui, Y. Du, R. Meka, A. Alaeddini, C. Li, S. Shirinkam, J. Wang, Development of a Deep Learning Model for Dynamic Forecasting of Blood Glucose Level for Type 2 Diabetes Mellitus: Secondary Analysis of a Randomized Controlled Trial, JMIR Mhealth Uhealth 7 (11) (2019) e14452. [Online]. Available: URL: https://mhealth.jmir.org/2019/11/e14452.
[33] E.M. Aiello, G. Lisanti, L. Magni, M. Musci, C. Toffanin, Therapy-driven deep glucose forecasting, Engineering Applications of Artificial Intelligence 87 (2020) 103255. [Online]. Available: URL: https://www.sciencedirect.com/science/article/pii/S0952197619302313.
[34] I. Fox, L. Ang, M. Jaiswal, R. Pop-Busui, J. Wiens, Deep multi-output forecasting: Learning to accurately predict blood glucose trajectories, in: Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining, July 2018. [Online]. Available: URL: https://doi.org/10.1145/3219819.3220102.
[35] A. Bertachi, L. Biagi, I. Contreras, N. Luo, J. Vehi, Prediction of blood glucose levels and nocturnal hypoglycemia using physiological models and artificial neural networks, in: CEUR Workshop Proceedings, vol. 2148, 2018, pp. 85–90. [Online]. Available: URL: http://ceur-ws.org/Vol-2148/paper14.pdf.
[36] C. Zecchin, A. Facchinetti, G. Sparacino, C. Cobelli, Jump neural network for online short-time prediction of blood glucose from continuous monitoring sensors and meal information, Computer Methods and Programs in Biomedicine 113 (1) (2014) 144–152. [Online]. Available: URL: https://www.sciencedirect.com/science/article/pii/S0169260713003234.
[37] J. Reifman, S. Rajaraman, A. Gribok, W.K. Ward, Predictive monitoring for improved management of glucose levels, Journal of Diabetes Science and Technology 1 (4) (2007) 478–486, pMID: 19885110. [Online]. Available: doi: 10.1177/193229680700100405.
[38] G. Sparacino, F. Zanderigo, S. Corazza, A. Maran, A. Facchinetti, C. Cobelli, Glucose concentration can be predicted ahead in time from continuous glucose monitoring sensor time-series, IEEE Transactions on Biomedical Engineering 54 (5) (2007) 931–937. [Online]. Available: URL: https://ieeexplore.ieee.org/document/4155016.
[39] E. Georga, V. Protopappas, D. Polyzos, D. Fotiadis, A predictive model of subcutaneous glucose concentration in type 1 diabetes based on random forests, in: 2012 Annual International Conference of the IEEE Engineering in Medicine and Biology Society, 2012, pp. 2889–2892. [Online]. Available: URL: https://ieeexplore.ieee.org/document/6346567.
[40] S. Hochreiter, J. Schmidhuber, Long short-term memory, Neural Computation 9 (8) (1997) 1735–1780. [Online]. Available: doi: 10.1162/neco.1997.9.8.1735.
[41] W.L. Clarke, D. Cox, L.A. Gonder-Frederick, W. Carter, S.L. Pohl, Evaluating clinical accuracy of systems for self-monitoring of blood glucose, Diabetes Care 10 (5) (1987) 622–628. [Online]. Available: URL: https://care.diabetesjournals.org/content/10/5/622.
[42] C. Marling, R. Bunescu, The ohiot1dm dataset for blood glucose level prediction: Update 2020, in: CEUR Workshop Proc., vol. 2675, 2020, pp. 71–74. [Online]. Available: URL: https://sites.google.com/view/kdhd-2018/.
[43] S. Taieb, A. Atiya, A bias and variance analysis for multistep-ahead time series forecasting, IEEE Transactions on Neural Networks and Learning Systems 27 (2016) 62–76. [Online]. Available: URL: https://ieeexplore.ieee.org/document/7064712.
[44] F.M. Bianchi, S. Scardapane, S. Løkse, R. Jenssen, Reservoir computing approaches for representation and classification of multivariate time series, arXiv e-prints, p. arXiv:1803.07870, Mar. 2018. [Online]. Available: URL: https://ui.adsabs.harvard.edu/abs/2018arXiv180307870B.

[45] F.M. Bianchi, S. Scardapane, S. Lokse, R. Jenssen, Reservoir computing approaches for representation and classification of multivariate time series, arXiv preprint arXiv:1803.07870, 2018. [Online]. Available: URL: https://arxiv.org/pdf/1803.07870.pdf.

[46] A. Oord, S. Dieleman, H. Zen, K. Simonyan, O. Vinyals, A. Graves, N. Kalchbrenner, A. Senior, K. Kavukcuoglu, WaveNet: A Generative Model for Raw Audio, arXiv e-prints, p. arXiv:1609.03499, Sept. 2016. [Online]. Available: URL: https://ui.adsabs.harvard.edu/abs/2016arXiv160903499V.

[47] I. Sutskever, O. Vinyals, Q.V. Le, Sequence to sequence learning with neural networks, 2014. [Online]. Available: URL: https://arxiv.org/abs/1409.3215.

[48] P.M. Desai, M.E. Levine, D.J. Albers, L. Mamykina, Pictures worth a thousand words: Reflections on visualizing personal blood glucose forecasts for individuals with type 2 diabetes, in: Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems CHI 18, 2018, pp. 1–13. [Online]. Available: doi: 10.1145/3173574.3174112.